

Assignment 14 Spring 2025

Heat Equation 1

BAR UNDER ADIABATIC CONDITIONS

“Adiabatic” means no heat exchange with the neighborhood, because the bar is completely insulated, also at the ends. *Physical Information:* The heat flux at the ends is proportional to the value of $\partial u / \partial x$ there.

11. Show that for the completely insulated bar, $u_x(0, t) = 0$, $u_x(L, t) = 0$, $u(x, t) = f(x)$ and separation of variables gives the following solution, with A_n given by (2) in Sec. 11.3.

$$u(x, t) = A_0 + \sum_{n=1}^{\infty} A_n \cos \frac{n\pi x}{L} e^{-(cn\pi/L)^2 t}$$

Find the temperature in Prob. 11 with $L = \pi$, $c = 1$, and $f(x) = \cos(2x)$.

Heat Equation 2

Solution to the heat equation where the ends of the bar are kept at 0°C is given by

$$u(x, t) = \sum_{n=1}^{\infty} u_n(x, t) = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{L} e^{-\lambda_n^2 t} \quad \left(\lambda_n = \frac{cn\pi}{L} \right)$$

With $u(x, 0) = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{L} = f(x)$, where $B_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx$.

Write down the solution $u(x, t)$ when $f(x) = \sin(0.1\pi x)$ and $c = 1$, $L = 10$.

Heat Equation 3

Solve the heat equation for a very long bar using Fourier transforms and the method of convolution to show that

$$u(x, t) = (f * g)(x) = \frac{1}{2c\sqrt{\pi t}} \int_{-\infty}^{\infty} f(p) \exp \left\{ -\frac{(x-p)^2}{4c^2 t} \right\} dp.$$

Hint: This is exactly what we did in class, so you just need to go through the notes.